THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 12: **Quadrilaterals** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** no three collinear, all in one plane, and non-self-intersecting sides	Definition 1 of the chapter
2	**(b)** has one internal angle greater than 180°	like the 'DART' shape
3	**(b)** a parallelogram	Theorem 5
4	**(a)** 2 : 1	longer part touches the vertex
5	**(d)** any quadrilateral, even a non-convex one	rotate a copy by 180° about the midpoint of a side
6	**(b)** Both A and R are true but R is not the correct explanation of A	both true; the Midpoint Theorem gives both facts, but being parallel does not explain the half-length
7	**(a)** 55°	adjacent angles of a parallelogram add up to 180°
8	**(a)** 4.5 cm	PQ = BC/2
9	**(d)** 16 cm	AG : GD = 2 : 1, so AG = 2/3 of AD
10	**(b)** 30 cm	each side of the medial triangle is half of a side

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	If three vertices are collinear, two adjacent sides lie along one line (1), giving a triangle or a figure with overlapping sides instead of a quadrilateral (1).	1+1
12	A quadrilateral all of whose internal angles are less than 180° (1). Test: its two diagonals intersect (1).	1+1
13	$\angle B = 180^\circ - 72^\circ = 108^\circ$ (1); $\angle C = 72^\circ$, $\angle D = 108^\circ$ (opposite angles equal) (1).	1+1
14	Converse: if the diagonals of a quadrilateral bisect each other, then it is a parallelogram (1). Yes, it is true (Theorem 4) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	In parallelogram ABCD, AC is a transversal for $AB \parallel DC$ and $AD \parallel BC$ (1). $\angle BAC = \angle DCA$ and $\angle BCA = \angle DAC$ (alternate angles), AC common ⇒ $\triangle ACD \cong \triangle CAB$ (ASA) (1). Hence $AB = DC$ and $AD = BC$ (1).	1+1+1
16	In quadrilateral ABCD, $AB = DC$, $AD = BC$ and AC is common ⇒ $\triangle ACD \cong \triangle CAB$ (SSS) (2). Corresponding angles are equal, so with transversal AC we get $AB \parallel DC$ and $AD \parallel BC$ (1).	2+1
17	$\angle A = \angle C = x$, $\angle B = \angle D = y$; angle sum $= 2(x + y) = 360^\circ$ (1) ⇒ $x + y = 180^\circ$, i.e. $\angle A + \angle B = 180^\circ$ and $\angle B + \angle C = 180^\circ$ (1). Co-interior angles add to 180° ⇒ $BC \parallel AD$ and $AB \parallel DC$ (1).	1+1+1
	OR: Diagonals AC and BD meet at E with $EA = EC$, $EB = ED$ (1). $\angle AED = \angle CEB$ (vertically opposite) ⇒ $\triangle AED \cong \triangle CEB$ (SAS) (1). So $\angle DAE = \angle ECB$ ⇒ $AD \parallel BC$; similarly $AB \parallel DC$ (1).	1+1+1
18	Let $AB \parallel DC$ and $AB = DC$, with diagonals meeting at E (1). $\triangle EAB \cong \triangle ECD$ (ASA: alternate angles and $AB = DC$) (1). So $EA = EC$, $EB = ED$: the diagonals bisect each other, so ABCD is a parallelogram (1).	1+1+1
	OR: $AD \parallel BC$ ⇒ $\angle DAC = \angle ACB$; $AB \parallel DC$ ⇒ $\angle BAC = \angle ACD$ (alternate angles) (2). As $\angle BAC = \angle DAC$, we get $\angle ACB = \angle ACD$, so AC bisects $\angle C$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) AC is a transversal of $AB \parallel DC$, $AD \parallel BC$ ⇒ $\triangle ACD \cong \triangle CAB$ (ASA) ⇒ $AB = DC$, $AD = BC$ (2). (b) $AD \parallel BC$ with transversal AB: $\angle A + \angle B = 180^\circ$; likewise any two adjacent angles add to 180° ⇒ $\angle A = \angle C$, $\angle B = \angle D$ (2). (c) With diagonals meeting at E, $\triangle AED \cong \triangle CEB$ (ASA) ⇒ $EA = EC$, $EB = ED$ (1).	2+2+1
	OR: (a) $AB = DC$, $AD = BC$, AC common ⇒ $\triangle ACD \cong \triangle CAB$ (SSS) ⇒ $\angle DAC = \angle ACB$, $\angle DCA = \angle CAB$ ⇒ $AD \parallel BC$, $AB \parallel DC$ (3). (b) $EA = EC$, $EB = ED$, $\angle AED = \angle CEB$ ⇒ $\triangle AED \cong \triangle CEB$ (SAS) ⇒ $AD \parallel BC$ (1); similarly $\triangle EAB \cong \triangle ECD$ ⇒ $AB \parallel DC$ (1).	3+1+1

20	(a) Angle sum: $2(\angle A + \angle B) = 360^\circ \Rightarrow \angle A + \angle B = 180^\circ = \angle B + \angle C \Rightarrow BC \parallel AD$ and $AB \parallel DC$ (3). (b) $AB \parallel DC$, $AB = DC$, diagonals meet at E: $\triangle EAB \cong \triangle ECD$ (ASA) ⇒ diagonals bisect each other ⇒ parallelogram (2).	3+2
	OR: Statement (1). Proof: draw $CR \parallel BA$ meeting PQ produced at R; $\triangle APQ \cong \triangle CRQ$ (AAS) ⇒ $CR = AP = BP$ (1); BCRP is a parallelogram ⇒ $PQ \parallel BC$ and $PQ = \frac{1}{2}PR = \frac{1}{2}BC$ (2). $PQ = 13$ cm (1).	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET B

Chapter 12: **Quadrilaterals** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** CD and ∠C	opposite sides share no vertex
2	**(a)** one pair of opposite sides equal	equal sides alone do not guarantee a parallelogram (an isosceles trapezium also has them)
3	**(a)** parallel to the third side and half of it	Theorem 6
4	**(d)** parallelogram	Varignon's theorem
5	**(a)** Both A and R are true and R is the correct explanation of A	Theorem 4
6	**(b)** 115°	adjacent angles of a parallelogram add up to 180°
7	**(a)** 9 cm	PQ = BC/2
8	**(c)** 14 cm	BC = 2 PQ
9	**(c)** 12 cm	AG = 2·GD, so AD = AG + GD = 3·GD
10	**(a)** 24 cm	its sides are AC/2, BD/2, AC/2, BD/2

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A quadrilateral with one pair of opposite sides both equal and parallel is a parallelogram (2).	2
12	$BC = 2 \times 7.5 = 15$ cm (1); $DE \parallel BC$ (Midpoint Theorem) (1).	1+1
13	$AG : GD = 2 : 1$ (1); $AG = 14$ cm and $GD = 7$ cm (1).	1+1
14	Sides are $\frac{9}{2}, \frac{13}{2}, \frac{9}{2}, \frac{13}{2}$ cm (1); perimeter $= 9 + 13 = 22$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\angle BAC = \angle DAC = \angle ACB$ (alternate angles) (1). In $\triangle ABC$, $\angle BAC = \angle BCA \Rightarrow AB = BC$ (1). Adjacent sides equal in a parallelogram ⇒ all four sides equal: a rhombus (1).	1+1+1
16	Statement: the segment joining the midpoints of two sides of a triangle is parallel to the third side and half as long (1). Proof: in $\triangle ABC$ with P, Q midpoints of AB, AC, draw $CR \parallel BA$ meeting PQ produced at R; $\triangle APQ \cong \triangle CRQ$ (AAS) ⇒ $CR = AP = BP$ (1). So BCRP is a parallelogram (equal parallel sides) ⇒ $PQ \parallel BC$ and $PQ = \frac{1}{2}PR = \frac{1}{2}BC$ (1).	1+1+1
17	Let P be the midpoint of AB and $PQ \parallel BC$ meet AC at Q. Let M be the midpoint of AC (1). By the Midpoint Theorem $PM \parallel BC$ (1). Only one line through P is parallel to BC, so PM and PQ coincide and $Q = M$ (1).	1+1+1
	OR: P, Q, R, S are the midpoints of AB, BC, CD, DA. In $\triangle ABC$: $PQ \parallel AC$; in $\triangle ADC$: $SR \parallel AC \Rightarrow PQ \parallel SR$ (2). Similarly $QR \parallel PS$ (using BD) ⇒ PQRS is a parallelogram (1).	2+1
18	$\angle A + \angle B = 180°$: $5x + 20 = 180 \Rightarrow x = 32$ (1). $\angle A = 111°$, $\angle B = 69°$ (1); $\angle C = 111°$, $\angle D = 69°$ (1).	1+1+1
	OR: Sum of ratio parts $= 12 \Rightarrow$ one part $= 30°$ (1). Angles: $60°, 90°, 90°, 120°$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Statement (1). Proof: let M be the midpoint of AC; PM $\parallel BC$ (Midpoint Theorem) (1); through P only one line is parallel to BC, so PQ and PM coincide and Q = M (2). $PQ = \frac{1}{2}BC = 15$ cm, $AQ = 11$ cm (1).	1+3+1
	OR: Let medians BQ, CP meet at M; X, Y the midpoints of BM, CM (1). $PQ \parallel BC$, $PQ = \frac{1}{2}BC$ and $XY \parallel BC$, $XY = \frac{1}{2}BC$ (Midpoint Theorem in $\triangle ABC$ and $\triangle MBC$) (1). So $\triangle MPQ \cong \triangle MYX$ (ASA) ⇒ $MQ = MX = XB$ and $MP = MY = YC$ (1). Hence $CM : MP = 2 : 1 = BM : MQ$ (1). The same argument for BQ and AR gives the same point and ratio ⇒ concurrent (1).	1+1+1+1+1
20	P, Q, R, S midpoints of AB, BC, CD, DA: in $\triangle ABC$ and $\triangle ADC$, $PQ \parallel AC \parallel SR$; in $\triangle BCD$ and $\triangle BAD$, $QR \parallel BD \parallel PS$ ⇒ parallelogram (3). If $AC = BD = 16$ cm all sides equal 8 cm: a rhombus (1), perimeter 32 cm (1).	3+1+1

	OR: PQRS is a parallelogram (Varignon) (1); PR and QS are its diagonals, so they bisect each other (2). If AC $=$ BD then $PQ = \frac{1}{2}AC = \frac{1}{2}BD = QR$, so PQRS is a rhombus (1); the diagonals of a rhombus are perpendicular ⇒ $PR \perp QS$ (1).	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND